

THE PATRON INDEX

Where Buyers Become Insights

Secure Customer Intelligence Platform

Global Innovation Hackathon 2026 • Team CodeGoblins

The Problem

Businesses have data—but not enough actionable customer intelligence.

Data Overload

Large volumes of purchase data are collected, but customer behaviour is difficult to interpret.

Missed Segments

Without effective segmentation, loyal, high-value, inactive and at-risk customers can remain hidden.

Lost Opportunities

Ineffective marketing and retention decisions can lead to missed revenue opportunities.

Our Solution

The Patron Index turns historical purchase data into customer segments and business insights.

- Securely validate uploaded business data before processing.
- Analyze purchase frequency, total spending, average order value and recent activity.
- Automatically classify customers into five meaningful segments.
- Present classifications, analytics, trends and recommendations in an interactive dashboard.

Customer Intelligence

Five segments make customer behaviour easier to understand and act on.

High Value Customers

Strong value contribution

Loyal Customers

Consistent engagement

Regular Customers

Steady purchasing behaviour

At-Risk Customers

Signs of declining activity

Inactive Customers

Low or no recent activity

How It Works

From raw purchase history to actionable customer intelligence.

01

Upload

Business purchase data

02

Secure

Validation + malware detection

03

Analyze

Purchase behaviour features

04

Classify

Customer segmentation

05

Visualize

Dashboard + trends

06

Act

Marketing & retention insights

Security by Design

Uploaded business data passes through a security layer before processing.

- File validation before downstream processing.
- Malware detection as part of the security module.
- Validation helps reduce unsafe or malformed input entering the pipeline.
- Security is integrated into the customer-intelligence workflow rather than treated as an afterthought.

Technology Stack

Modular architecture for frontend, backend, ML and security.

Frontend

React.js • Tailwind CSS •
Chart.js

Backend

Node.js • Express.js

Database

MongoDB / MySQL

Machine Learning

Python • Pandas •
Scikit-learn

ML API

FastAPI

Security

Malware Detection • File
Validation

Version Control

Git • GitHub

Business Value

The platform connects customer data with practical business decisions.

- Marketing: identify customer groups for more targeted engagement.
- Retention: surface at-risk customers for timely intervention.
- Customer engagement: understand purchasing patterns and recent activity.
- Decision support: combine classifications, analytics, trends and recommendations in one dashboard.

What Makes the Product Different

A single workflow combines intelligence, visualization and input security.

Customer Segmentation

Moves beyond raw transaction tables by translating behaviour into five understandable customer categories.

Actionable Dashboard

Presents classifications, analytics, trends and recommendations for business decision-making.

Security Layer

Validates uploaded data and incorporates malware detection before processing.

Roadmap & Closing

Build → validate → demonstrate → scale.

- Phase 1: Secure data upload and validation pipeline.
- Phase 2: Customer feature engineering and segmentation.
- Phase 3: Interactive analytics dashboard and recommendations.
- Phase 4: API integration, testing, deployment and refinement.
- Vision: help businesses turn purchase history into clearer customer intelligence.

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